LECTURE 4

Wireless Network Security

This lecture addresses the security of wireless networks. It begins with why wireless security is essential and introduces the three main concepts of authentication, encryption, and integrity. It then reviews the foundational ideas on which wireless security depends: network layers, 802.11 networks, and the basic cryptographic tools. Next it presents authentication methods, ranging from the least secure to the most secure methods used in modern networks, followed by the encryption and integrity methods TKIP, CCMP, and GCMP. The lecture closes with Wi-Fi Protected Access (WPA, WPA2, and WPA3), the standard sets of security protocols developed by the Wi-Fi Alliance.

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4.1 Introduction to Wireless Network Security

4.1.1 Why Wireless Security Is Essential

Security is important in all networks; however, it is even more essential in wireless networks. The main reason is that **wireless signals are not contained within a wire**: any device within range of the signal can receive the traffic.

In wired networks, traffic is often encrypted only when it is sent over an untrusted network such as the Internet, and wired traffic within the LAN is usually not encrypted. In wireless networks, however, it is critical to encrypt traffic sent between wireless clients and the access point (AP), because any device within range can receive it. The goal is to ensure that only the intended recipient or recipients can read the traffic.

Three main concepts are covered in this lecture: **authentication**, **encryption**, and **integrity**.

4.1.2 Authentication

Authentication means verifying the identity of a user and/or device. All clients must be authenticated before they can associate with an AP.

The AP should authenticate the identity of clients, but ideally clients should also authenticate the AP, to avoid associating with a malicious AP. A malicious AP could trick users into associating with it and then carry out attacks such as a man-in-the-middle attack.

Authentication can be carried out in several ways: with a password, with a username and password combination, or with digital certificates installed on the devices. Before associating with an AP, the client must first go through an authentication process.

4.1.3 Encryption

Encryption means scrambling a message so that only the sender and the intended recipient can read it. Traffic sent between clients and APs (in fact, any wireless traffic) should be encrypted so that it cannot be read by anyone except the AP and the client.

Many protocols can be used to encrypt traffic, but the sender and recipient must use the same protocol; otherwise, the recipient would not be able to decrypt the sender's messages.

4.1.4 Integrity

Integrity ensures that a message is not modified by a third party: the message sent by the source host should be the same as the message received by the destination host. To ensure this, a **MIC (Message Integrity Check)** is added to messages. The process works as follows.

1. **Calculate the MIC.** The sender calculates a MIC for the message and attaches it to the message. Many protocols can be used to calculate the MIC, and the sender and receiver must use the same protocol.
2. **Encrypt and send.** The sender encrypts the message and the MIC and sends the frame.
3. **Decrypt and recalculate.** The recipient decrypts the message and then independently calculates a MIC for the message, using the same protocol as the sender.
4. **Compare.** The recipient compares the two MICs. If the MIC calculated by the sender and the MIC calculated by the recipient are the same, the recipient assumes that the message was not tampered with.

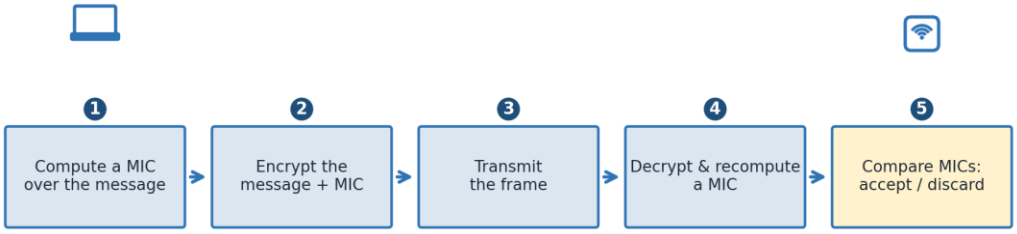


Figure 4.1 *How a Message Integrity Check (MIC) is generated, transmitted, and verified*

If the two MICs are different, the message is discarded. Rather than saying that a MIC protects the integrity of messages, it is more accurate to say that it helps identify whether the integrity of a message has been compromised; if it has, the message is discarded.

4.2 Foundational Concepts: Layers, 802.11 Networks, and Cryptography

Wireless security is built from a small set of ideas that are recombined throughout this lecture. This section reviews where wireless security sits in the network layers, how 802.11 networks are organized, and the handful of cryptographic tools on which authentication, encryption, and integrity depend.

4.2.1 Wireless Security at Layer 2

Networking is described in layers, where each layer does one job and does not care about the details of the others. **Authentication, encryption, and integrity all happen at or just above**

Layer 2, which is why security can be changed independently of the rest of the network stack (Table 4.1).

Table 4.1 *The network layers relevant to wireless security*

Layer	Name	Job
Layer 1	Physical	Sends and receives the actual radio waves
Layer 2	Data Link	Packages data into frames; uses hardware addresses to decide which local device a frame is for
Layer 3	Network (IP)	Routes packets between networks

Two key definitions are used throughout the lecture.

- **Frame:** the basic unit of data sent and received at Layer 2. Every "frame" in this lecture (an authentication request, an EAP message, an encrypted data frame) is one of these units.
- **MAC (Media Access Control) address:** a 48-bit hardware address (for example, 00:1A:2B:3C:4D:5E) burned into every network interface and used at Layer 2 to deliver frames to a specific device.

Layer 3 and above are completely unaware of which wireless security protocol is used: a strongly encrypted Wi-Fi frame and a weakly encrypted one carry IP packets in exactly the same format.

4.2.2 Wireless Networking Basics

An **802.11 network** is the technical name for Wi-Fi: a set of IEEE standards for wireless local area networks (WLANs) in which devices communicate using radio waves instead of cables. This lecture focuses on **infrastructure mode**, in which every client communicates through an AP. The main network structures are as follows.

- **BSS (Basic Service Set):** an AP plus its associated clients, identified by the AP's radio MAC address, called the **BSSID**.
- **SSID (Service Set Identifier):** the human-readable network name that users see in the Wi-Fi list. The BSSID is hidden from users.
- **ESS (Extended Service Set):** several BSSs that share one SSID and are connected by a common distribution system, so that a client can roam between APs without reconnecting manually.
- **Infrastructure mode:** every client communicates through an AP. This is the normal case and is assumed throughout the lecture.
- **IBSS / ad hoc mode:** clients communicate directly, with no AP. It is rare in managed environments and is not covered further.

A MAC address is a 48-bit number, typically written as six pairs of hexadecimal digits. It is burned into the hardware by the manufacturer and is intended to be globally unique. When a frame arrives at a device, the network card checks whether the destination MAC address matches its own; if not, it ignores the frame (unless it is a broadcast). In wireless security, the BSSID is simply the MAC address of the AP's radio, a crucial identifier in network management.

4.2.3 Cryptography Primer: Ciphers, Keys, and Certificates

4.2.3.1 Symmetric and Asymmetric Encryption

Symmetric encryption uses one shared secret key, whereas asymmetric (public-key) encryption uses a mathematically related pair of keys: a public key that anyone may have and a private key that only its owner has (Table 4.2).

Table 4.2 *Symmetric and asymmetric encryption compared*

	Symmetric	Asymmetric (public-key)
Keys	One shared secret key (or one derived from it)	A mathematically related pair: public key (anyone) and private key (owner only)
Speed	Fast	Far slower
Used for	Bulk data encryption (WEP, TKIP, CCMP, GCMP)	Briefly: authenticating identities and establishing a shared symmetric key
Example use	Encrypting the actual wireless traffic	Inside the TLS tunnels used by PEAP and EAP-TLS

The pattern is consistent: wireless protocols use slow asymmetric cryptography only long enough to authenticate and agree on a symmetric key, and then switch to fast symmetric encryption for the actual data. This is exactly what happens inside the TLS tunnels of PEAP and EAP-TLS.

4.2.3.2 Block Ciphers and Stream Ciphers

- A **block cipher** (for example, AES) encrypts fixed-size chunks of data and needs a mode of operation to handle messages of arbitrary length.
- A **stream cipher** (for example, RC4) generates a continuous pseudo-random keystream, which is combined with the plaintext one bit at a time using XOR.

XOR outputs 1 if two bits differ and 0 if they match ($0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, $1 \oplus 1 = 0$). Its useful property is that XOR-ing the same value twice cancels out: plaintext $\oplus$ keystream = ciphertext, and ciphertext $\oplus$ keystream = plaintext. This is exactly how the receiver of a stream cipher recovers the original data, using only the same keystream that the sender used.

The two kinds of cipher appear in the wireless protocols as follows.

- **WEP and TKIP** use RC4, a stream cipher, directly.
- **CCMP and GCMP** use AES, a block cipher, in Counter mode, which makes it behave like a stream cipher.

4.2.3.3 Digital Signatures

A digital signature is the mechanism on which a certificate relies. It works in three steps.

1. **Hash the document.** The signer hashes the document to produce a short fingerprint.
2. **Encrypt the fingerprint.** The signer encrypts the fingerprint with their own private key, which only they can do.
3. **Verify.** Anyone with the signer's public key decrypts the fingerprint, independently hashes the document, and compares the two.

If the two match, the document must have been signed by the holder of that private key and has not been altered since. Note the reversal: ordinary encryption uses the **recipient's public key** (so that only the recipient can decrypt), whereas a signature uses the **signer's own private key** (so that anyone can verify who signed).

4.2.3.4 Digital Certificates and PKI

A **Certificate Authority (CA)**, a trusted third party, digitally signs a document that binds a public key to an identity, such as a company's RADIUS server. Anyone who trusts that CA can verify the signature and be confident that the public key really belongs to that server, without ever having met it. This CA-based trust structure is a **Public Key Infrastructure (PKI)**. It is exactly what a PEAP or EAP-TLS client relies on when it verifies the authentication server's certificate before handing over any credentials.

4.2.3.5 Hash Functions and MICs

A hash function and a MIC both produce a short value from a message, but they protect against different things (Table 4.3).

Table 4.3 *A hash function compared with a MIC*

	Hash function	MIC (Message Integrity Check)
Secret key?	No: anyone can recompute it	Yes: it folds in a shared secret key
Protects against	Accidental corruption only	Deliberate tampering by an attacker
Why	An attacker can simply recompute a matching hash after tampering	Only someone who knows the key can produce a valid MIC

> **IMPORTANT DISTINCTION**
> In general cryptography, a MIC is sometimes called a MAC (Message Authentication Code). This is completely unrelated to a MAC address. 802.11 documentation uses "MIC" specifically to avoid this confusion.

4.2.4 Keys: Unicast and Group

Wireless networks use two kinds of key so that unicast traffic and broadcast traffic can be protected separately.

- **Pairwise key:** each associated client negotiates its own unique key with the AP. This ensures that other clients on the same WLAN cannot decrypt one another's unicast traffic, even though everyone uses the same encryption protocol.
- **Group key:** a shared key held by all clients, distributed and periodically refreshed by the AP. It is used only for broadcast and multicast traffic, when the AP needs to send one frame to every client at once.

> **THE IMPORTANT POINT**
> Wireless security is a Layer 2 concern, operating over 802.11 infrastructure networks (BSS/ESS), and it exists because anyone in range can receive the signal. Its tools are symmetric ciphers (RC4 stream cipher for WEP and TKIP; AES block cipher in Counter mode for CCMP and GCMP) for bulk encryption; asymmetric cryptography and certificates (PKI), used briefly to authenticate identities and establish symmetric keys inside TLS tunnels; pairwise and group keys to separate unicast from broadcast traffic; and keyed MICs, not plain hashes, to detect deliberate tampering. The radio standard (Wi-Fi 4, 5, 6, or 7) and the security configuration are independent layers.

4.3 Authentication Methods

This section provides an overview of seven authentication methods, ranging from the least secure to the most secure. The first two, open authentication and WEP, are the two options included in the original 802.11 standard, and neither is secure. The remaining five, the EAP framework and four EAP methods (LEAP, EAP-FAST, PEAP, and EAP-TLS), are more secure methods that were developed afterwards. A basic understanding of each is sufficient.

4.3.1 Open Authentication

The original 802.11 standard included two options for authentication: **open authentication** and **WEP** (described in section 4.3.2). The first is **open authentication**: the client sends an authentication request and the AP accepts it, with no questions asked and no credentials required. This is not a secure authentication method, because the AP accepts all authentication requests. Authentication takes place before association: the client sends an authentication request and receives a response, and only then sends an association request (Figure 4.2).



Figure 4.2 *The 802.11 authentication and association handshake*

However, it is still used today in combination with other authentication methods. After the client is authenticated and associated with the AP, the user may be required to authenticate by other methods before access to the network is granted. Open authentication itself is not secure, but it can be combined with other methods.

PRACTICAL EXAMPLE

Public Wi-Fi networks often allow devices to associate with the AP without a password, but then require the user to log in through a web page. After that authentication, the device is granted access to the Internet.

4.3.2 WEP (Wired Equivalent Privacy)

The second authentication option in the original 802.11 standard is **WEP**. WEP is not just an authentication method; it also provides encryption. For encryption, WEP uses the **RC4** algorithm. WEP is a **shared-key** protocol: the sender and receiver must have the same key. WEP keys can be 40 bits or 104 bits in length, and they are combined with a 24-bit initialization vector to bring the total length to 64 bits or 128 bits.

Longer key lengths are usually more secure, but **WEP is not secure and can be easily cracked regardless of the key length**. WEP should not be used on modern wireless networks.

WEP can also be used to authenticate a device. The AP is testing whether the client knows the shared key (Figure 4.3).

1. **Challenge.** The AP sends a challenge phrase, a series of bits whose actual contents do not matter.
2. **Response.** The client encrypts the challenge phrase using the WEP key and sends it back.
3. **Comparison.** The AP compares the client's encrypted challenge phrase with its own encrypted challenge phrase. If they match, the AP and client are using the same key, and authentication is successful.

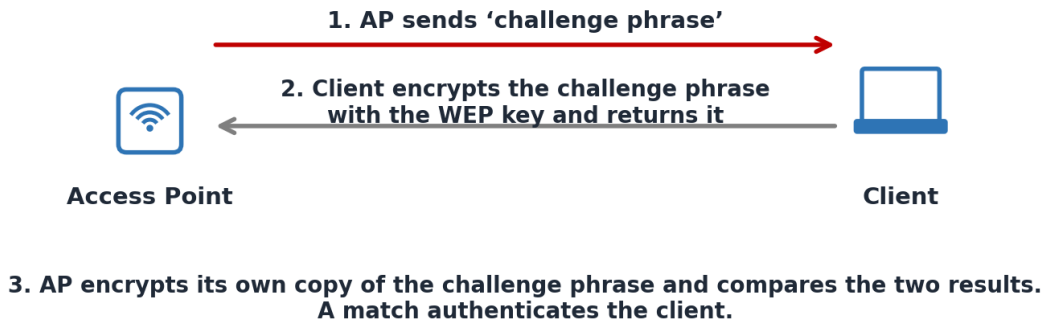


Figure 4.3 *WEP's challenge-response authentication exchange*

WEP can be used just to provide encryption, or to provide both encryption and authentication. If WEP authentication is not used, open authentication is used and WEP serves only for encryption. Open authentication on its own is not secure, and WEP is not secure either, as an encryption method or as an authentication method. More secure wireless authentication methods were therefore developed.

4.3.3 EAP (Extensible Authentication Protocol)

EAP is not one of the two original 802.11 options, and it is not a type of WEP. **EAP** is not a single authentication protocol but a framework on which many other protocols, called **EAP methods**, are based, such as LEAP and EAP-FAST. EAP defines a standard set of authentication functions

used by the various EAP methods. Four of those methods are covered here: LEAP, EAP-FAST, PEAP, and EAP-TLS.

4.3.3.1 EAP and 802.1X

EAP is integrated with a protocol called **802.1X**, which provides port-based network access control. 802.1X limits network access for clients connected to a LAN or wireless LAN until they authenticate, and it is used in all kinds of networks, both wired and wireless. There are three main entities in 802.1X, and these three definitions are essential.

1. **Supplicant.** The device that wants to connect to the network.
2. **Authenticator.** The device that provides access to the network.
3. **Authentication server (AS).** The device that receives the client's credentials and permits or denies access.

In an 802.11 wireless LAN, the supplicant is the client that wants to connect (for example, a laptop). The authenticator is the device that provides access to the network, which is the access point. In a split-MAC architecture, the wireless LAN controller (WLC) manages the authentication, not the AP itself. The authentication server is usually a **RADIUS** server.

The 802.11 authentication required to associate with the AP is simply open authentication. The client is then associated but does not yet have full access to the network, and the only traffic allowed from the client is the traffic required for EAP authentication. The authentication server checks the client's credentials and decides whether to permit or deny access to the network, with the WLC acting as a middleman in the process. The 802.11 authentication that is needed simply to associate with the AP is therefore open, but an additional step of EAP authentication is required to gain access to the network (Figure 4.4).

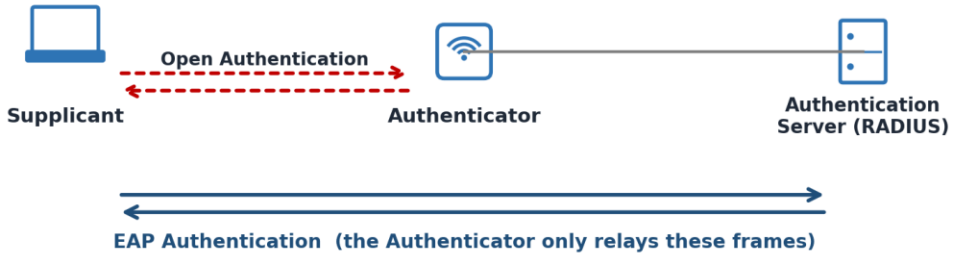


Figure 4.4 *The 802.1X entities and the two-stage flow: open authentication for association, followed by EAP authentication for network access*

4.3.3.2 LEAP (Lightweight EAP)

LEAP was developed by Cisco as an improvement over WEP. Clients must provide a username and password to authenticate, and mutual authentication is provided by both the client and the server sending a challenge phrase to each other. In WEP, only the server sends the challenge phrase; in LEAP, both client and server do. Each device encrypts the other's challenge phrase and sends it back, and uses the result to authenticate the other, as in WEP (Figure 4.5).

To further improve security, **dynamic WEP keys** are used: WEP keys that automatically change over time to make the encryption harder to crack. Like WEP, however, LEAP is considered vulnerable and should not be used. One of the following methods should be used instead.



Figure 4.5 *LEAP's mutual challenge exchange between client and server, relayed through the AP*

4.3.3.3 EAP-FAST (EAP Flexible Authentication via Secure Tunnel)

EAP-FAST was also developed by Cisco. It consists of three phases (Figure 4.6).

1. **Generate and pass the PAC.** A **PAC (Protected Access Credential)** is generated and passed from the server to the client. The PAC is like a shared key and is used in the next phase.
2. **Establish the tunnel.** The PAC is used to establish a secure TLS tunnel between the client and the authentication server, which can be used for secure, encrypted communications.
3. **Authenticate the client.** The client is authenticated within that TLS tunnel.

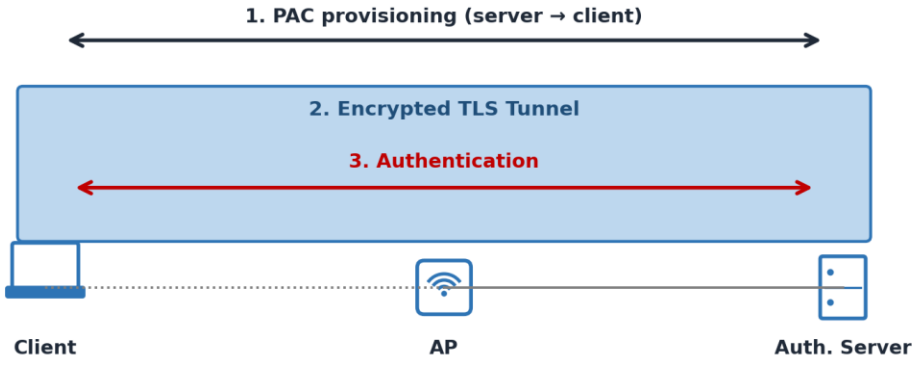


Figure 4.6 *EAP-FAST's three phases: PAC provisioning, TLS tunnel establishment, and authentication inside the tunnel*

4.3.3.4 PEAP (Protected EAP)

PEAP is similar to EAP-FAST but differs in key ways. Like EAP-FAST, it involves establishing a secure TLS tunnel between the client and the server. Instead of a PAC, however, the server has a **digital certificate**. It shows the certificate to the client, and the client uses it to authenticate the server. The same certificate is also used to establish the secure TLS tunnel (Figure 4.7).

Because only the server provides a certificate, the client must still be authenticated within the secure tunnel. One protocol that can be used for that authentication is **MS-CHAP** (Microsoft Challenge Handshake Authentication Protocol). In short, EAP-FAST uses a PAC, whereas PEAP uses a digital certificate; both establish a secure tunnel and then authenticate the client inside it.

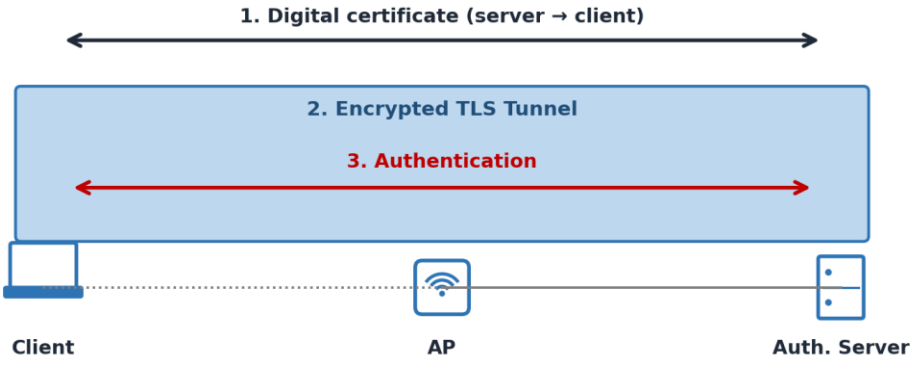


Figure 4.7 *PEAP: the server's digital certificate establishes the TLS tunnel; the client authenticates inside it*

4.3.3.5 EAP-TLS (EAP Transport Layer Security)

PEAP requires only the authentication server to have a certificate. **EAP-TLS** requires a certificate on the authentication server and on every single client. Because the client and server authenticate each other with digital certificates, there is no need to authenticate the client within the TLS tunnel; the tunnel is still used, however, to exchange encryption key information (Figure 4.8).

EAP-TLS is considered the **most secure** authentication method, but it is more difficult to implement than PEAP, because every client device needs a certificate, which adds complexity, time, and effort. Many enterprises therefore prefer PEAP instead.

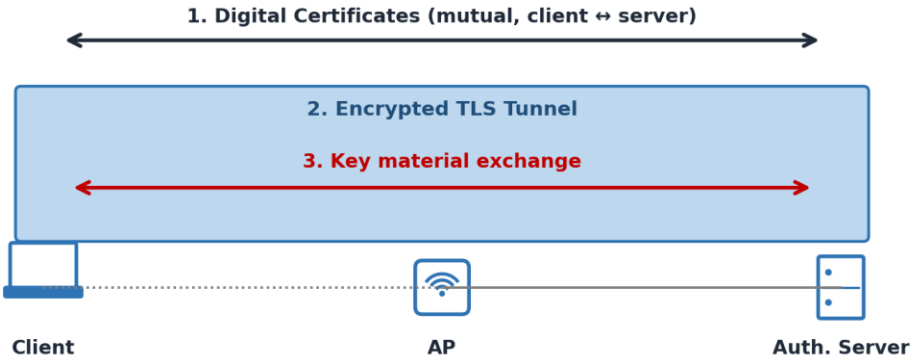


Figure 4.8 *EAP-TLS: both client and server present digital certificates to establish a mutually authenticated TLS tunnel*

4.3.3.6 Comparison of the EAP Methods

Table 4.4 *Summary of the EAP methods*

Method	Developer	How the server is authenticated	How the client is authenticated	Status
LEAP	Cisco	Challenge phrase (mutual)	Username and password, with a challenge phrase	Vulnerable; should not be used
EAP-FAST	Cisco	PAC, used to build a TLS tunnel	Inside the TLS tunnel	Secure
PEAP	–	Server digital certificate, used to build a TLS tunnel	Inside the TLS tunnel, for example with MS-CHAP	Secure; easier to deploy than EAP-TLS
EAP-TLS	–	Server digital certificate	Client digital certificate; no inner authentication needed	Most secure; hardest to deploy

Authentication refers to verifying the identity of a device or user. The next section addresses the specific encryption methods used to keep data private and secure.

4.4 Encryption and Integrity Methods

The encryption methods discussed in this section are TKIP, CCMP, and GCMP. WEP, introduced earlier, is also used for integrity. Encryption of wireless traffic is essential, and a basic understanding of these methods is required.

4.4.1 TKIP (Temporal Key Integrity Protocol)

WEP was found to be vulnerable, but the wireless hardware of the time was built to use WEP, so a temporary solution based on WEP was needed until a new standard was created and new hardware was built. That temporary solution was **TKIP**. It is based on WEP but adds several security features, which makes it a more secure version of WEP.

- TKIP uses a **MIC** to protect the integrity of messages.
- A **key mixing algorithm** creates a unique WEP key for every frame, instead of every frame using the same key, which makes the encryption harder to crack.
- The **initialization vector is doubled** from 24 bits to 48 bits, which makes brute-force attacks on the encryption much more difficult.
- The MIC includes the **sender MAC address**, which is used to prove the identity of the frame sender.
- A **timestamp** is added so that attackers cannot perform a replay attack, which involves resending frames that have already been sent.
- A **TKIP sequence number** keeps track of the frames sent from a source MAC address. This also protects against replay attacks, because a frame is dropped if the sequence counter is incorrect.

TKIP is used in WPA (Wi-Fi Protected Access version 1).

4.4.2 CCMP (Counter Mode CBC-MAC Protocol)

CCMP was developed after TKIP and is more secure. It is used in WPA2. For a device to use CCMP, the device's hardware must support it: old hardware built only for WEP or TKIP cannot use CCMP.

CCMP consists of two algorithms, one for encryption and one for a message integrity check. For encryption it uses **AES counter mode**. AES is the most secure encryption protocol currently available and is used throughout the world by corporations, governments, and other organizations. AES has multiple modes of operation, and CCMP uses counter mode because it offers high performance and speed. As the message integrity check, CCMP uses **CBC-MAC** (Cipher Block Chaining Message Authentication Code), a kind of MIC.

4.4.3 GCMP (Galois Counter Mode Protocol)

GCMP is more secure and more efficient than CCMP, and its efficiency allows higher data throughput. It is used in WPA3. Like CCMP, GCMP consists of two algorithms: first **AES counter mode** encryption, and then **GMAC** (Galois Message Authentication Code), which is used as a MIC to ensure the integrity of messages. GMAC is a kind of MIC.

4.4.4 Comparison of the Methods

Table 4.5 *Encryption and integrity methods*

Protocol	Encryption	Integrity check	Used in
TKIP	RC4-based (WEP with per-frame key mixing and a 48-bit IV)	MIC (includes the sender MAC address)	WPA
CCMP	AES counter mode	CBC-MAC	WPA2
GCMP	AES counter mode	GMAC	WPA3

4.5 Wi-Fi Protected Access: WPA, WPA2, and WPA3

With so many different authentication, encryption, and integrity methods, which ones should be chosen? Which combinations work together, and which do not? How can it be determined which hardware supports which standards? To simplify this and create standard sets of protocols, the Wi-Fi Alliance developed **WPA (Wi-Fi Protected Access)**.

4.5.1 WPA Certifications Overview

The Wi-Fi Alliance has developed three WPA certifications for wireless devices: **WPA**, **WPA2**, and **WPA3**. The first is called WPA, not WPA1. For a device to be WPA certified, it must be tested in authorized testing labs. The Wi-Fi Alliance certifies devices for Wi-Fi 4, Wi-Fi 5, Wi-Fi 6, and other standards, and it also certifies devices for WPA, WPA2, and WPA3 security.

4.5.2 Authentication Modes: Personal and Enterprise

The WPA certifications support two authentication modes.

1. **Personal mode.** A **pre-shared key (PSK)** is used for authentication. Entering the password when connecting to a home Wi-Fi network is personal mode, which is common in small networks such as SOHO networks. For security, the PSK itself is not sent over the air: a **four-way handshake** is used for authentication, and the PSK is used to generate encryption keys. If two devices use the same PSK to generate encryption keys, they can decrypt each other's traffic, which shows that the other device is using the same PSK.

2. **Enterprise mode.** This mode uses **802.1X with an authentication server**, and the EAP methods introduced earlier are used with it. WPA does not specify any particular method, so all EAP methods are supported, such as PEAP or EAP-TLS.

4.5.3 WPA (Wi-Fi Protected Access)

The first WPA certification was developed after WEP was proven to be vulnerable. It includes the following protocols.

- **TKIP**, which is based on WEP, provides encryption and integrity.
- Authentication is provided by **802.1X and EAP** in Enterprise mode, or by **PSK** in Personal mode.

WPA did not last long. After more secure protocols and hardware were developed, WPA2 was released.

4.5.4 WPA2 (Wi-Fi Protected Access 2)

WPA2 includes the following protocols.

- **CCMP** is used to provide encryption and the MIC.
- Authentication can be done via **802.1X using EAP**, or by **PSK**.

4.5.5 WPA3 (Wi-Fi Protected Access 3)

WPA3 was released in 2018. It uses **GCMP** for encryption and integrity and supports 802.1X-based or PSK-based authentication. WPA3 also offers several additional security features.

- **PMF (Protected Management Frames):** protects 802.11 management frames from eavesdropping and forging. This was available as an optional feature in WPA2, but WPA3 makes it mandatory.
- **SAE (Simultaneous Authentication of Equals):** protects the four-way handshake when Personal mode authentication is used.
- **Forward secrecy:** prevents data from being decrypted after it has been transmitted over the air. This protects against attacks in which an attacker captures wireless frames and then tries to decrypt them later to read the content.

4.5.6 Comparison of WPA, WPA2, and WPA3

Table 4.6 *The three WPA certifications*

	WPA	WPA2	WPA3
Encryption and integrity	TKIP	CCMP (AES counter mode with CBC-MAC)	GCMP (AES counter mode with GMAC)
Personal mode	PSK	PSK	PSK, with SAE protecting the four-way handshake
Enterprise mode	802.1X with EAP	802.1X with EAP	802.1X with EAP
Protected management frames	Not available	Optional	Mandatory
Forward secrecy	No	No	Yes